1. Sliding-Window Context at Step t

Folded Action: history (1..t-4) + Active visual window (t-3, t-2, t-1) → Target (Step t)

System & Task

- GUI + Bash tools
- Task instruction
- 3-part format

Folded Steps (1 .. t-4)

Screenshots dropped

Action 1: Open VS Code
Action 2: Write swatch.ppm

Long-horizon memory

Active Visual Window (3 screenshots + full responses)

Turn t-3	Turn t-2	Turn t-1 (Now)
Shot t-3	Shot t-2	Shot t-1
<think> + Action: + Call	<think> + Action: + Call	<think> + Action: + Call

Supervised Target (Step t)

1. <think> rationale
2. Action: summary
3. <tool_call>

2. History Window Comparison

10 Text + 1 Shot ... × Blind to no-op clicks

Cannot compare before/after screens → repeats failed clicks

Folded + 3 Shots Action 1..t-4 ✓ Sees state changes

3 shots verify UI transitions + Action: log tracks prior steps

3. Step-Level Rationale Synthesis

Task & prior actions

Before / After shots

Executed <tool_call>

→ **Synthesis Pass**

Grounds CoT & Action: summary

→

1. First-person <think>
2. Imperative Action:
3. Verified <tool_call>